

Research paper

Real-time structural validation for material extrusion additive manufacturing

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ABSTRACT

Material extrusion additive manufacturing is a technology that produces a part by controlling the melting and extrusion of a thermoplastic filament. A void defect in a specific position can significantly impact the whole product's structural quality and mechanical properties. Different defect detection systems have been built and have achieved good detection accuracy. However, a challenge remains in determining if the structural performance of a part with defects will fail to meet design requirements. This research develops a real-time product structural quality validation system using a multi-dimensional accumulation-threshold-based decision-making approach. The proposed system is validated on a consumer-grade 3D printer with an optical camera. The layer-wise damage information is output from a trained convolutional neural network. The developed structural quality validation system links the obtained defect information with the decision boundary developed by the accumulation-threshold-based decision-making approach to evaluate the effect of the defect. The accumulation-threshold-based decision-making approach is trained on a novel component health index that links the location and size of defects across layers. Experimental results show that the decision boundary obtained from the accumulation-threshold-based decision-making approach performs well on test data with 96% recall and a 91% F1-score, which means the decision boundary can provide good overall results while being biased to limit false negative results. The proposed real-time structural quality validation system is validated online for a dog-bone test specimen. Results show that the computational time of the structural quality validation system requires at most 688 ms, while a dog-bone layer takes 75 s to print, thus demonstrating that the proposed system can meet required real-time constraints. The capability to run online enables users to cancel a print mid-process for specimens that will not meet structural loading requirements. The developed system is demonstrated through a video, which is provided in the supplemental materials. The dataset for this work is published as a public repository containing 450 samples with 221 failure classes.

1. Introduction

Material extrusion additive manufacturing (MEXAM), also known as fused filament fabrication or fused deposition modeling, is one of the fastest-growing, most promising, and widely-used additive manufacturing technologies [1]. The unique affordability of MEXAM has made it a popular technique for small-scale 3D printing. However, its wide-scale adoption for industrial applications still faces significant challenges, especially for Industry 4.0, due to the process's intrinsic uncertainties [2,3]. MEXAM is a repetitive process: the molten material squeezes from the nozzle to form a thin, tacky bead of plastic that adheres and bonds with the previous layer until the part is finished. During the printing process, defects, such as bonding gaps and/or material voids, can dramatically decrease the final product's structural quality and mechanical properties [4,5]. Therefore, a defect detection

system for MEXAM that reduces part-to-part variations is essential to ensure the product's mechanical properties.

Machine learning (ML) is an effective way to detect defects or guarantee product quality for MEXAM due to its ability to link implicit information with the final prediction or classification outcome [6]. Extensive ML systems for MEXAM have been built to detect various faults, including bonding quality problems and infill defects [7]. For example, Jin et al. used an optical camera mounted on a material extrusion printer to capture different product quality images for under-extrusion, good-quality, and over-extrusion conditions [8]. These images were used to build a dataset, which was used to train a ResNet 50 architecture convolutional neural network (CNN) classification algorithm. Online monitoring was implemented by feeding the captured images into the pre-trained classification model to get the current printing

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condition. Banadaki et al. built a reliable online quality monitoring system with a deep convolutional neural network (DCNN) algorithm [9]. After training the model with images captured at each layer, the proposed online model can classify five different quality grades resulting from overfilled or underfilled with an average accuracy of 94%. Wu et al. presented an approach to detect malicious infill defects injected into the 3D printing process using a classification-based ML method to analyze images from a camera [10]. The achieved infill defect prediction accuracies are 85.26% and 95.51% with a naive Bayes classifier and a J48 decision tree, respectively. Holzmond et al. presented a MEXAM product quality assurance system that uses a three-dimensional digital image correlation system to capture the geometry and then compare the obtained geometry with the computer model to achieve in situ printing error detection [11]. A validation experiment was done on a MEXAM printer, which demonstrates the in situ detection of localized and global defects. An in situ infrared temperature sensing for real-time fused deposition modeling defect detection was studied by Hossain et al. [12]. A statistical method for defect detection is exploited and applied to detect temperature deviations, which are caused by deliberately rooted defects. Khan et al. utilized a CNN algorithm to model to detect real-time malicious defects in the printing process, which is based on feature extraction of geometrical anomalies occurring in infill patterns [13]. Charalampous et al. presented a vision-based real-time monitoring of MEXAM for automatic manufacturing error detection [14]. The proposed method compared the high-resolution point cloud data of the printed part with the corresponding digital 3D model in various phases of the printing process to automatically monitor and detect the defect. Cases validate the efficacy of the proposed methodology.

The ultimate goal of defect detection is to guarantee product quality rather than just find defects. Degradation in functional qualities can be caused by the accumulation of smaller defects that, on their own, do not constitute a critical flaw in the component. Therefore, an effective online quality validation system must consider the accumulation of smaller defects when attempting to guarantee product quality. To expand, the proposed system should not only detect visible defects but also diagnose their direct and compound effects on product functional qualities (e.g., structural performance) caused by variations in the printing process (e.g., abnormal extrusion).

In this paper, a real-time ML-based product structural quality validation approach is proposed. The novel method is capable of inferring structural quality rather than just detecting surface defects and uses an accumulation-threshold-based decision-making system to account for the accumulation of smaller defects. If a single defect or the accumulation of smaller defects has a limited effect on product quality, this means the defect is ignorable, and the printing can continue. Otherwise, the printing process needs to stop once the defect is impactful. In addition, a novel component health index that links the size of defects across layers is developed in this work. The component health index accumulates defect information throughout the whole sample printing process. The ML-based product defect detection approach uses the CNN algorithm to classify the defect type. Tensile tests are done on the collected samples with different defects to get the effect-of-defect on product quality. The decision boundary is built in the component health index domain using a support vector machine (SVM) with a radial basis function kernel trained against the ultimate tensile load obtained from mechanical testing. Layer-wise images are fed into the trained CNN model to classify defects during the printing process. Then the accumulated component health index is mapped to the decision boundary model to evaluate the effect-of-defect on product quality. Based on the assumption that all un-printed layers of the part are of good quality, the multi-dimensional accumulation-threshold-based decision-making system will make the proper decision (i.e., continue or cancel printing) by referring to the evaluation result before the next layer is finished. The real-time evaluation and decision-making

continue for every layer until the whole part is done or the impactful defect occurs.

The experimental setup and methodology are discussed in Section 2, which has five sections: experiment methodology, experiment setup, data-driven model for defect detection, component health index, and defect impact boundary. Section 3 discusses the experiment results and provides a discussion, which includes training results for the data-driven defect detection, results for the defect impact decision boundary process, and results of the real-time defect detection and decision-making process. Section 4 presents the conclusion and future work.

2. Experimental setup and methodology

2.1. Experiment methodology

The real-time structural quality validation system for MEXAM is diagrammed in Fig. 1. The system has three parts:

1. Offline defect detection training.
2. Offline decision boundary training.
3. Online real-time decision-making.

In the offline defect detection training process, a dataset is built that consists of images taken at each layer during manufacturing with an optical camera. Images are then labeled manually with the defect information (i.e., defect length and location). The dataset is used to train the CNN model. The trained CNN model is deployed to infer the corresponding defect length and location during the normal printing process in real-time. The data-driven ML model is discussed in greater detail in Section 2.3.

The decision boundary is trained offline on data collected from the tensile tests for the samples with defects to evaluate the effect-of-defect [15,16]. The defect is considered impactful when it reduces the structure's tensile strength by 3% [17], an arbitrary threshold based on the author's experience; and breaks at the seeded defect position. Otherwise, the defect is ignorable, which means the defect has a limited effect on product quality. The defect impact decision boundary is built using an SVM with the radial basis function kernel based on the tensile test results from the collected samples with various defects. The decision boundary lays the foundation for the accumulation-threshold-based effect-of-defect inference. The trained defect impact decision boundary is deployed for online real-time decision-making. A decision on the structural quality of the part is made for every layer in the build. At every layer, the inference from the CNN model will be accumulated with the previous information and then mapped to the defect impact decision boundary to evaluate the impact on product quality. Based on the evaluation result, the accumulation-threshold-based decision-making system will make the proper decision [18].

2.2. Experimental setup

Samples are printed with a printer (manufacturer: Creality, model: Ender 5, city: Shenzhen, country: China) with Polylactic Acid filament (manufacturer: Hatchbox, thickness: 1.75 mm, city: Los Angeles, country: United States). Fig. 2(a) presents the setup of the experimental platform. An optical camera (manufacturer: JAI, model: CV-M4+CL, city: Los Angeles, country: United States) with a resolution of 1368×1020 is fixed to the printer frame above the extruder to collect the printing images. The designed sample's G-code was modified before printing, so when one layer is completed, the printing head moves to a fixed position, which relocates out of the camera's field for the duration of the picture. At this time, the camera is triggered through a LabVIEW code to capture the layer image of the sample. With the setup, the relative position between the camera and specimens is fixed, providing consistent images with an identical scale across layers.

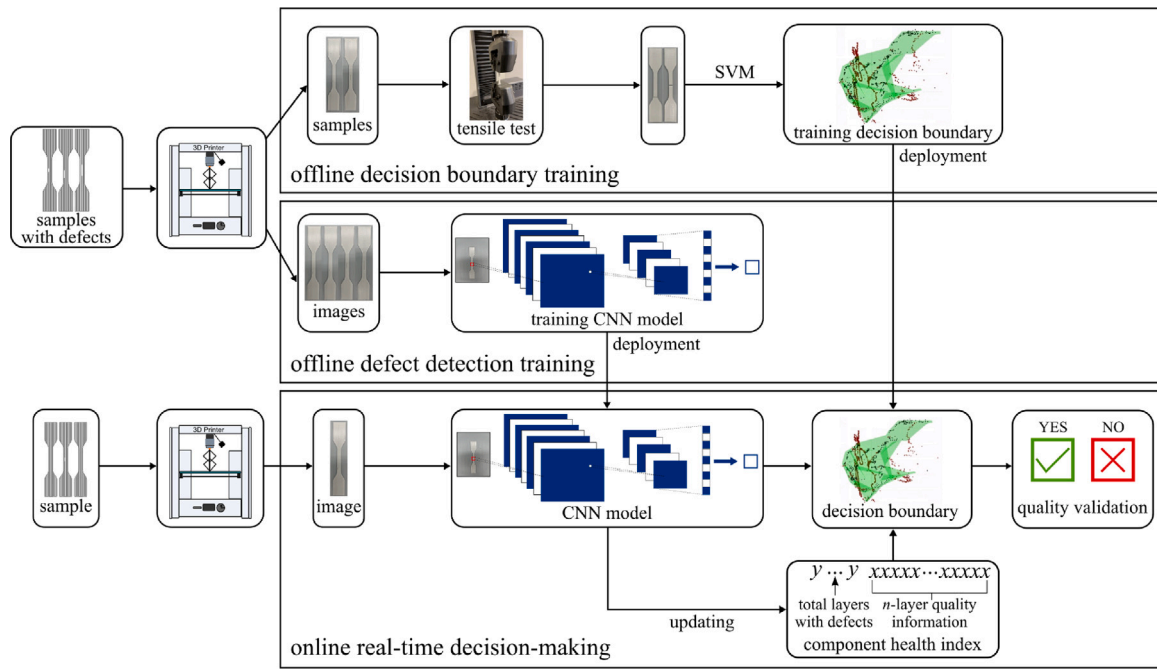


Fig. 1. Diagram for the real-time structural quality validation approach in the MEXAM printing process.

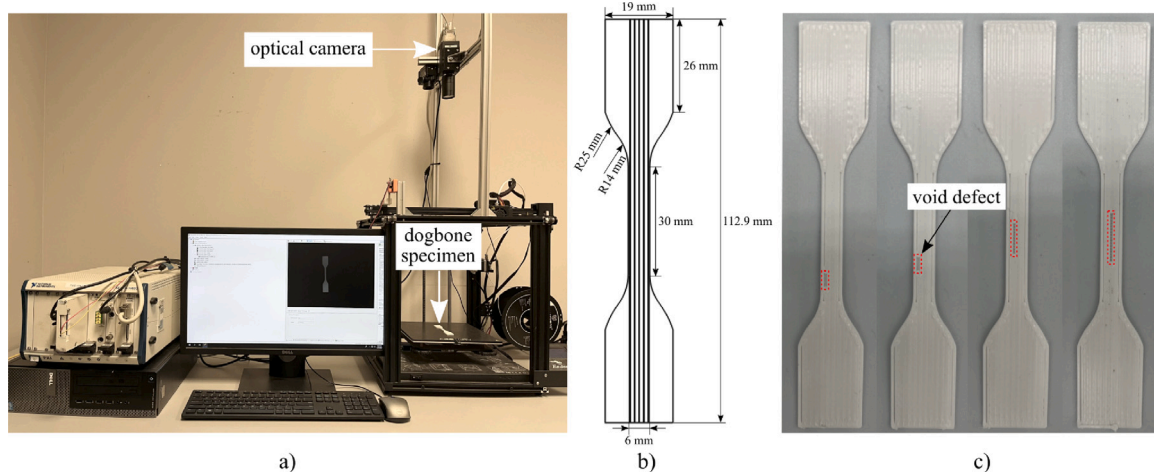


Fig. 2. The experimental platform and designed sample: (a) experimental platform for defect detection and product structural quality validation; (b) the designed sample dimension; and (c) various designed defects on real printed samples.

As MEXAM manufactures the part by depositing the plastic filament, each layer is bonded to the adjacent layers. The overlap regions (inter-layer necking) between the adjacent layers bond together through filament interface diffusion to produce a semi-solid part [19,20]. The voids between tracks introduced during the printing may be ignorable, which means they have a tolerable impact on the part's mechanical property. However, large gaps in the extruded polymer undermine the product's quality [21].

The defects in this research are produced by turning the extrusion off, which simulates the abnormality of the failure-to-feed during the printing process. The dimensions of the designed dog-bone are shown in Fig. 2(b), which is based on the American Society for Testing Materials (ASTM) D638 Type IV dog-bone test specimen [22]. The sample thickness is 3 mm with 100% infill. The diameter of the printing nozzle is 0.8 mm to accelerate printing speed and reduce the sample collection time. There are nine layers for the sample: the bottom and top layers are printed in the horizontal direction, and no defects happen on these two layers; the internal layers are printed in the vertical direction, and

all defects are introduced into these seven layers. As shown in Fig. 2(b), only five extrusion traces cross through the whole dog-bone, which are the actual extrusions that the defect occurs on. To increase the size of the training dataset, data augmentation (i.e., horizontal and vertical flipping, Gaussian blur, and Gaussian noise) is implemented.

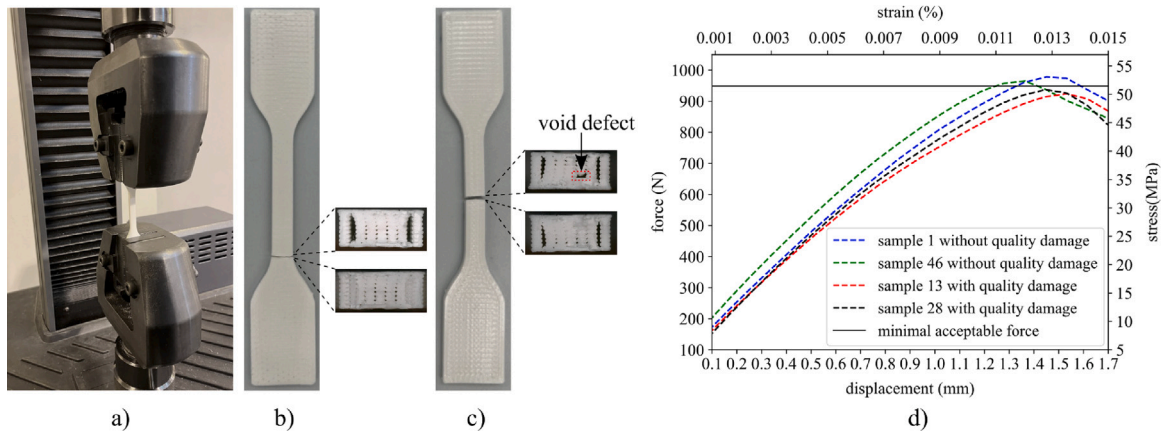
Based on finite element analysis of failures for the printing parts by Garg et al. the stress distribution on the sample printed by MEXAM varies in diverse locations in gauge length and layers [23]. Thus, the defect in different locations and layers may have varying impacts on product structural quality. To figure out the defect's effects on product structural quality, four defect variables are used in this study: defect length, defect location, defect occurring layer, and the number of layers with defects.

Table 1 reports the 30 designed defects with three defect lengths (5, 10, and 15 mm) used to investigate their impact on product quality. The x, y position is located at the center of the defect (Note that only the 30 mm straight traces cross through the whole dog-bone), which is used for calculating the defect location based on equation $\sqrt{x^2 + y^2}$. For

Table 1

The designed defect with different lengths and locations within the dog-bone specimen.

length location	5 mm				10 mm			15 mm		
x1	y1	y2	y3	y4	y5	y6	y7	y8	y9	y10

**Fig. 3.** Printed sample tensile test: (a) tensile test setup; (b) the broken good quality sample with its cross-section; (c) the broken sample with defect and its cross-section; and (d) the tensile force results.

the 5 mm defect, there are four y locations in the lower-left quadrant and one that crosses into the upper-left quadrant at one x location (x location is the traces cross through the entire dog-bone, which are termed line 1, line 2, and line 3). The defect on one line is either 2.5, 7.5, 12.5, or 15 mm in one of four different positions. Therefore, based on the equation above, the 5 mm defect has four potential locations (2.7, 7.6, 12.5, 15) on line 1. This style of defect is repeated for the 5, 10, and 15 mm defect, as shown in Table 1. Fig. 2(c) shows the different defect lengths in various locations. Note that even though the sample infill density is 100%, due to the big diameter of the nozzle, the slicing resolution is low, which produces the long void along the edge (the long voids shown in Fig. 3(b) and (c)).

Since the accumulated defects' impact on product structural quality is important to consider [24,25], two variables, the layer of defect occurrence and the number of layers with defects, are defined. The details are presented in Section 2.4.

To train the product structural quality defect impact decision boundary, tensile tests are done for the collected samples. Mechanical validation is performed using ASTM D638 on an electromechanical load frame (manufacturer: MTS, model: Exceed E43, city: Minneapolis, country: United States), as shown in Fig. 3(a). To have a thorough and accurate evaluation of defect impact, tensile force reduction with the rooted defect is considered [26]. The average tensile force for ten samples without seeded void defects is 977.9 N, with a standard deviation of 9.92 N. In addition, the average tensile force for ten good samples where images are not collected and therefore the layers are not allowed to cool during image acquisition is 982.4 N, with a standard deviation of 8.43 N. The tensile force for the good samples with and without image acquisition interval are within one standard deviation and therefore no significant difference is found. In this work, a 3% reduction in the ultimate force of the good sample with image capture is chosen as the minimal acceptable structural threshold [27,28]; this relates to a 29.34 N reduction in the tensile force at failure. For

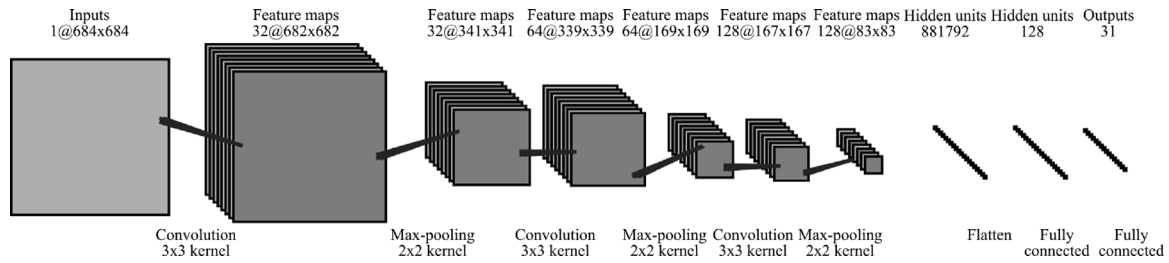


Fig. 4. Designed data-driven CNN model structure with three convolutional layers, three max-pooling layers, two fully connected hidden layers and one output layer.

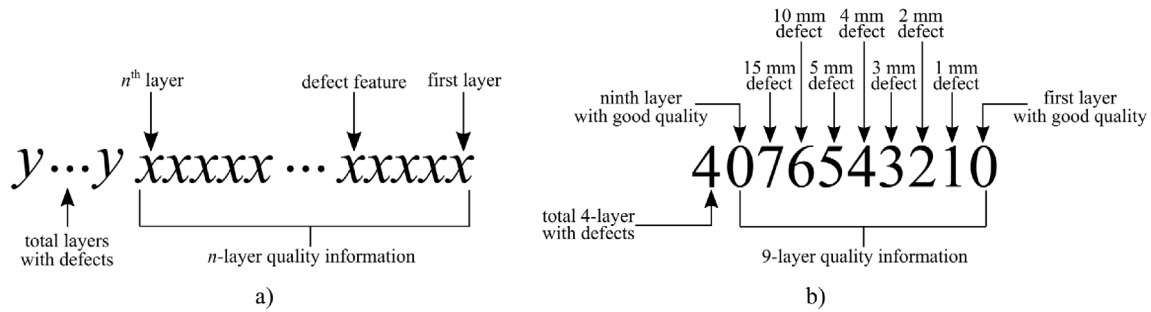


Fig. 5. The proposed component health index for recording the product layer quality: (a) the generalized component health index; and (b) a 9-layer component health index for the dog-bone specimen.

example, even if a sample breaks at the location of a defect but the tensile force is larger than 948.6 N, the defect's effect on the product's mechanical properties is still treated as ignorable due to the maximum tensile force having no meaningful difference. Therefore, the hypothesis in this research is that the defect is considered impactful if and only if the specimen breaks with a tensile force of less than 948.6 N at the location of a detected defect. Fig. 3(b) and (c) show the broken sample's cross-section after the mechanical testing. Fig. 3(b) displays the sample's broken cross-section without defect. Fig. 3(c) is the sample with a 10 mm defect in the center of the x_2 location; the break occurred at the defect (a red rectangle marks the void defect), which means this defect impacts the product's structural quality. Fig. 3(d) shows the tensile forces of the samples with and without quality damage. Note that all the tensile tests in the dataset are recorded with displacement and force. The dataset developed for this work is made available in a public repository [Dataset-in-situ-validation-for-polymer-components], where it is Dataset-1 [29].

2.3. Data-driven model for defect detection

The CNN model in this research is developed with three convolutional layers, three max-pooling layers, two fully connected hidden layers, and one output layer. A network similar to this was previously demonstrated by the authors to have excellent defect detection capabilities [30]. The designed model architecture is shown in Fig. 4 and is based on lenet5 [31]. The convolutional kernel size adopted in this model is 3×3 , and the filters are 32 with one stride. Except for a softmax activation function for the last output layer, all the rest are Rectified Linear Units (ReLUs). The raw image dimension of the sample is 1368×1020 . To increase the training speed, the image resolution is reduced by resizing to 684×684 pixels before being input into the model. After the first convolution operation on the resized input image, the output feature map size is 682×682 . A max-pooling layer with a 2×2 kernel size is performed after the first convolutional layer. The feature map reduces from 682×682 to 341×341 after downsampling. By performing the subsequent convolutional and max-pooling layers twice, the output feature map size changes to 83×83 . After flattening all the feature maps, two fully connected hidden layers

are used to depict the whole image's feature map by a one-dimensional vector, which dramatically reduces the size from 881792 to 128. The last output layer is also a fully connected layer with a total node of 31 (30 defect types with defect length and location, good quality). The output nodes represent the prediction categories from the algorithm, where each node maps to one defect type or good quality. The CNN model is trained offline using Keras and Tensorflow using the Adam optimizer [32]. After data augmentation, the final dataset contains 1550 images. For the labeled sample images, 80% of them are randomly chosen as training data, and the remaining 20% are used as test data. Training is done over 200 epochs. The defect length and defect location are output by the CNN model; however, with just an unlabeled image, the CNN cannot tell which layer has the defect and the number of layers with defects. This is the job of the component health index, which is discussed next.

2.4. Component health index

The proposed component health index is a standardized way to accumulate the effects of defects in the printed component while assuming that all future layers of the part will be of perfect quality. The developed component health index is applied to record product layer information, which contains two parts: which layer has a defect and the number of layers with defects. A graphical representation of the generalized component health index is presented in Fig. 5(a). Starting from the right, for a printed component with n layers, the first n digits are associated with an individual layer. Each layer quality written here as x , is quantified on a scale, which can be assigned to specific quality features. On the left, a single number represents the total number of layers with defects. This means that the number of y digits is defined as the total number of digits required to count the total number of layers. Mathematically, this is written as $\lfloor \log_{10} n \rfloor + 1$. The initial values of the component health index are all set to zero, which by default assumes all successive new layers are without defects.

Fig. 5(b) shows an example of the proposed component health index for the dog-bone specimen considered in this work. The dog-bone is a 9-layer print with seven defects. In this preliminary work, the first and last layers, corresponding to the bottom and top layers, are always

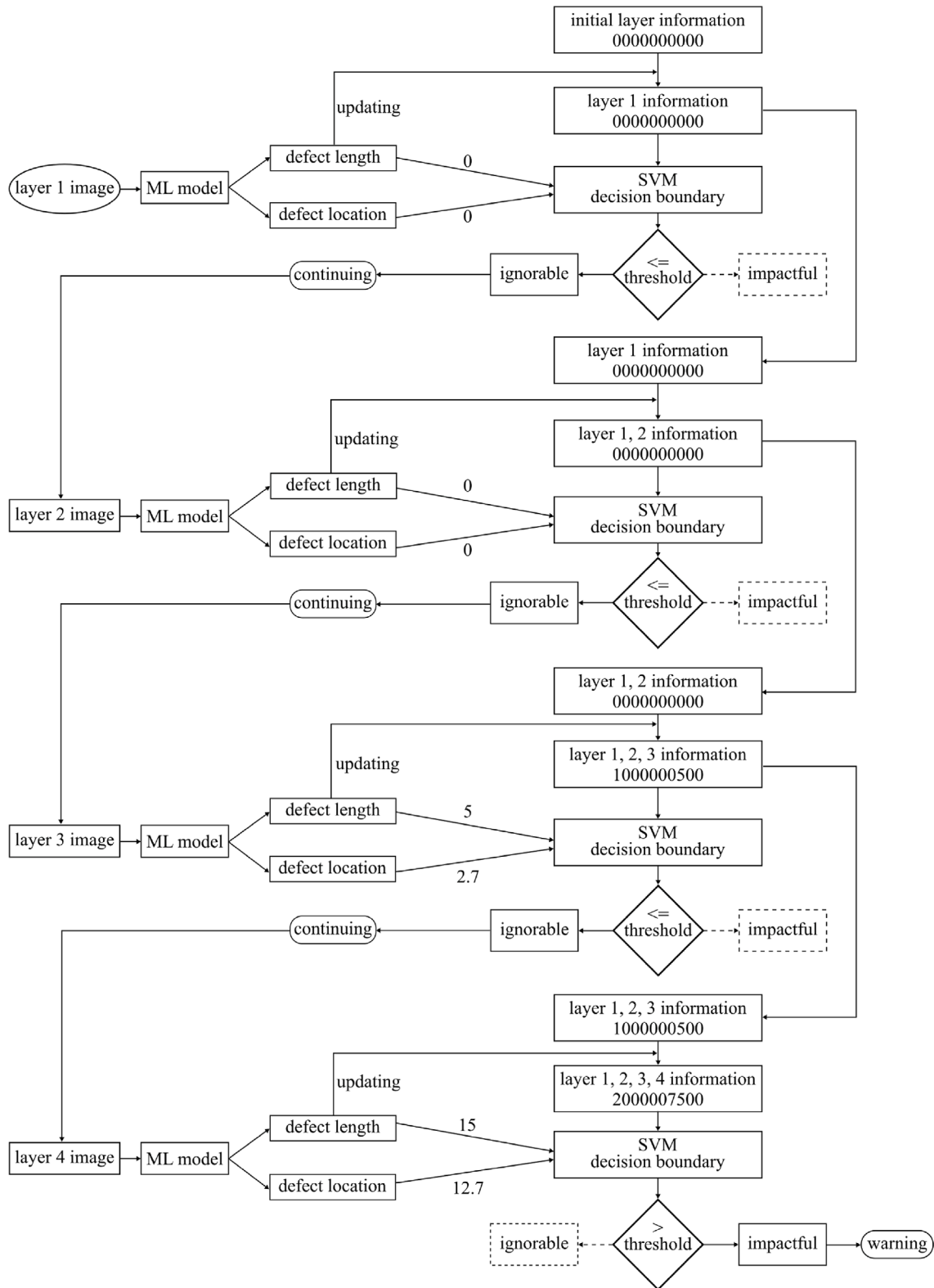


Fig. 6. Visualization of the structural validation and decision-making process.

considered to be without defects. Therefore, all defects occur in the seven infill layers. The component health index has seven damage quantization levels, which were designed for this research. 0 represents a good quality layer, while 1, 2, 3, 4, 5, 6, and 7 represent defects with lengths of 1, 2, 3, 4, 5, 10, and 15 mm, respectively. The generic component health index can be expanded to apply to any new cases.

2.5. Defect impact decision boundary

The total number of defect impact results from the tensile test is 19503; 80% are split as training data, and 20% are chosen as test data to plot the decision boundary. Of note is that assumptions based on the real defect impact result are used to expand the data obtained from

mechanical testing. To be more specific, suppose a 10 mm defect on one layer of the sample damages the product's structural quality and is broken at the rooted defect position. Therefore, a sample with more defects than just the 10 mm defect would also impair the product's structural quality by rooted defects. Three variables, defect length, defect location, and component health index, are chosen to draw the defect impact decision boundary. The SVM model with radial basis function kernel is trained offline using scikit-learn [33]. As the dataset is somewhat imbalanced, the weights of the SVM's classes are balanced to avoid overfitting. The defect impact decision boundary here is only for a quadrant of the dog-bone specimen, and the symmetry of the part is leveraged to expand the decision boundary for the entire part.

2.6. Real-time defect detection and decision-making

Real-time defect detection has two steps. First, an image is captured of the recently extruded layer and fed into the trained ML model. The model outputs the layer's quality information (i.e., defect length and defect location or good quality). Second, the health index is updated to combine the layer-related information (good quality or defect with length, location, and layer information) together. The updated health index is then mapped to the decision boundary for evaluation. If the accumulation of defects has a limited impact on structural quality impact and can be ignored, the printing and evaluation process will continue. However, suppose a combination of defects occurs such that the defect accumulation impact on the decision boundary shows that it is severe enough to undermine the product's structural quality. In that case, it may be prudent to cancel the print as the subsequent deposition of layers will not improve the quality of the part.

The SVM-based decision-making process uses a pre-computed boundary that is trained offline before printing is initiated. The current position of the part within the part's health performance space is computed using an SVM computation that uses input defect length, defect location, and the component health index. If the current position of the part's health marker is below the decision boundary, it is implied that the defect has a limited impact on structural quality and can be ignored. Otherwise, the product quality is undermined by the detected defects.

Fig. 6 shows a visualization of the structural validation and decision-making process. In the printing process, the first layer image is captured, and as it is the bottom layer and assumed to be of good quality in this introductory work, the component health index is not changed. Next, the second layer image is fed into the ML model. After obtaining the prediction result from ML, the quality of this layer is also good, therefore, the component health index remains unchanged. The printing continues to the third layer, with a 5 mm defect at position 2.7. The component health index is updated from 0000000000 to 1000000500. After evaluation against the decision boundary, the output result is below the threshold. Then the fourth layer image feeds into the ML model, and a 15 mm defect is detected at position 12.7 on this layer. After updating, the layer information is refreshed from 1000000500 to 2000007500, which means two layers have defects, a 5 mm defect on the third layer and a 15 mm defect on the fourth layer. By evaluating against the decision boundary function, it is determined that the value of the part's health marker is above the threshold value. This means the accumulated defects have an impactful effect on the product's structural quality. At this point, a warning for the part is provided. If deemed appropriate, the print may be canceled.

To ensure the proposed algorithm can satisfy the timing constraints for real-time product structural quality validation, an investigation of the algorithm's timing is undertaken. Fig. 7 reports a diagram of the computational cycle used for the real-time product structural quality validation. For this investigation, the total cycle time is studied; this includes: capturing the image; processing data through the CNN model; and performing the multi-dimensional accumulation-threshold-based decision-making processes. In this work, a personal computer with an i7-3770 CPU @ 3.40 GHz with 8 GB RAM was utilized.

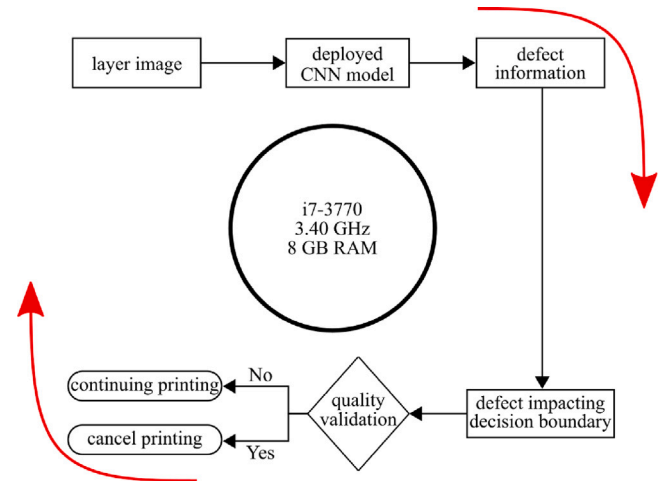


Fig. 7. The diagram of the time range calculation for the real-time product structural quality validation.

3. Results and discussions

3.1. Training results for the data-driven defect detection

The CNN for defect classification is the first step of the process, which largely determines whether the following structural quality validation and decision-making are successful. Therefore, the accuracy of the data-driven defect detection from CNN is an important evaluation factor. The ultimate result for training data achieves 100% accuracy, recall, and precision with a 100% macro-averaged F1-score (the average of each class's F1-score), as the acquisition conditions of the images and geometry of the samples are always almost the same. In addition, the unbalanced nature of the dataset plays a role in these high values. The algorithm also performs well on test data, with a 99.78% average accuracy, 99.57% average recall, and a 99.57% average precision. The macro-averaged F1-score for test data is 99.57%. The training and validation results verify that the designed CNN model is precise and can be used for MEXAM defect prediction. Note that these results are excellent due to the somewhat large errors considered here and the maturity of CNNs for fault detection on hand-labeled data. Moreover, CNN has strong robustness for image classification.

3.2. Result of defect impact decision boundary process

Fig. 8 reports a 3D view of the part's health performance space where the volume of the green shape is considered to be a non-impactful defect (most of the impactful space is removed to provide a clear view of the decision boundary). The green is the prediction decision boundary for the ignorable defects, which means if the defect is located in this area, it has a limited (i.e., ignorable) effect on product quality. All area outside of the green decision boundary is treated as impactful to product structural performance. Moreover, location 0, 0, 0 is a special location in the health performance space as it represents a part without detectable defects. Note that all three plots are different views of the same health performance space. In this 3D space, the x and y axis are defect length and defect location, while the z axis is the component health index. In the decision boundary plot, data denoted with a red cross means the defect is ignorable and the black dot means the defect is impactful to the product's structural performance.

Note that in Fig. 8, plenty of impactful defects are deep within the area defined by the decision boundary. However, two ignorable defects are located outside of the decision boundary. The reason is that the training of the algorithm done by the team emphasizes recall over precision, as the miss classification of a faulty component as functional

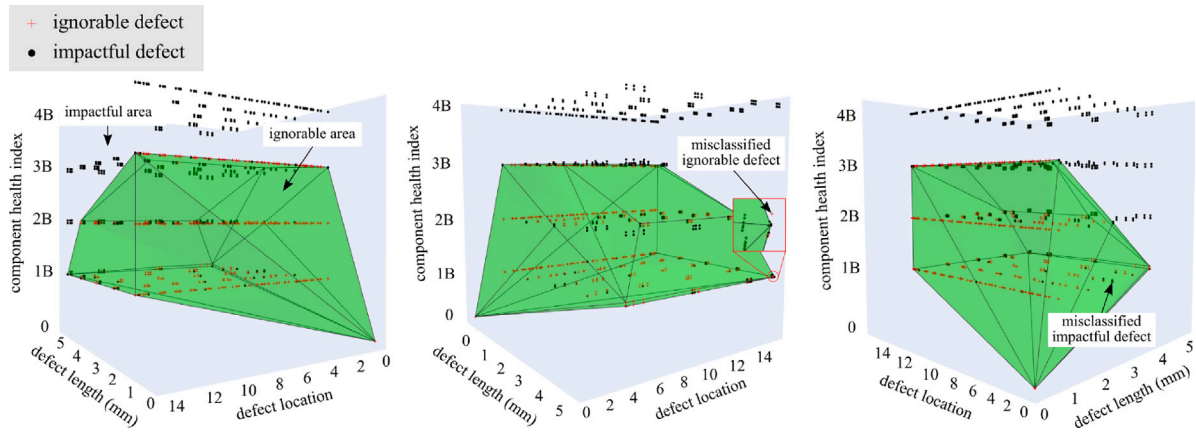


Fig. 8. Defect impact decision boundary from different views (the green is the prediction decision boundary for the defects, red crosses are observations of ignorable defects, and black dots are observations of impactful defects.). An interactive 3D version of the decision boundary is available in this manuscript's supplemental documents. In the interactive version, blue diamonds are correctly classified test data on the decision boundary and the pink squares are incorrectly classified test data on the decision boundary.

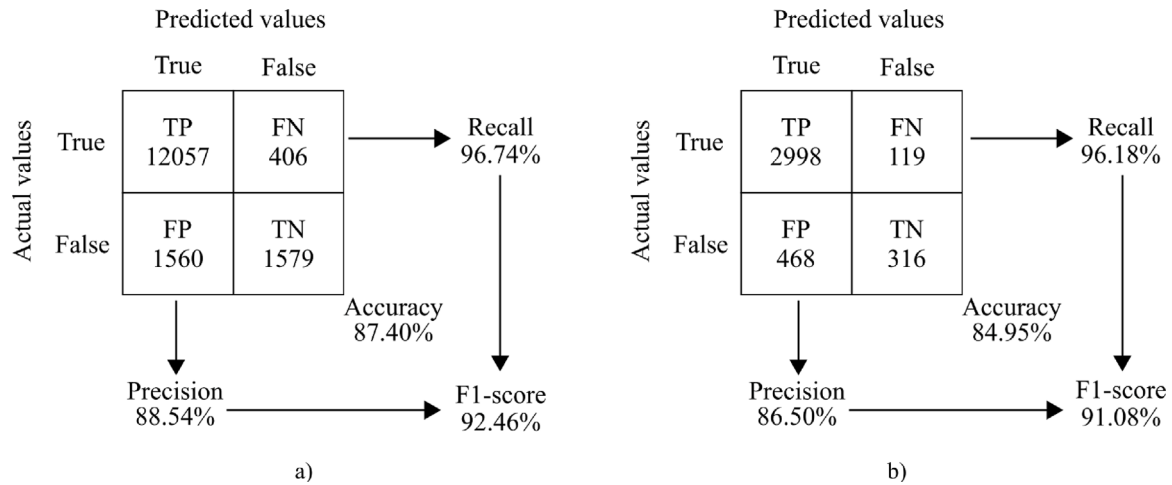


Fig. 9. The confusion matrices for the decision boundary, showing: (a) the confusion matrix for the training data; and (b) the confusion matrix for the test data.

is more detrimental than a functional component as faulty [34]. Recall that the project's hypothesis is that the defect is considered impactful if and only if the specimen breaks with a tensile force of less than 948.6 N at the location of a detected defect. The decision boundary confusion matrix on the training and test data can be constructed, as shown in Fig. 9 (a) and (b), respectively. The F1-score for the test data shows the decision boundary accurately predicts the defect's effect on product structural quality. Again, note that the algorithm is biased towards recall, with 96.18% for the test data.

For completeness, a true positive is a specimen that is predicted to have impactful defects and fails under the 948.6 N threshold at the location of a detected defect. A true negative is a specimen that is predicted to have ignorable defects and fails above the 948.6 N threshold at any location. A false positive is a specimen predicted to have impactful defects yet fails above the 948.6 N threshold at any location. Conversely, a false negative specimen is predicted to have ignorable defects yet fails below the 948.6 N threshold at any location.

The test data with 30 samples is used to validate the trained decision boundary (please refer to the online interactive 3D decision boundary). Of these thirty samples, twenty-five samples are predicted correctly. The interactive 3D decision boundary plot shows the correctly predicted test data as blue squares, while the five incorrectly predicted samples are plotted with pink squares. Table 2 details the

five incorrectly predicted test samples in an effort to investigate the reasons for the algorithm failures.

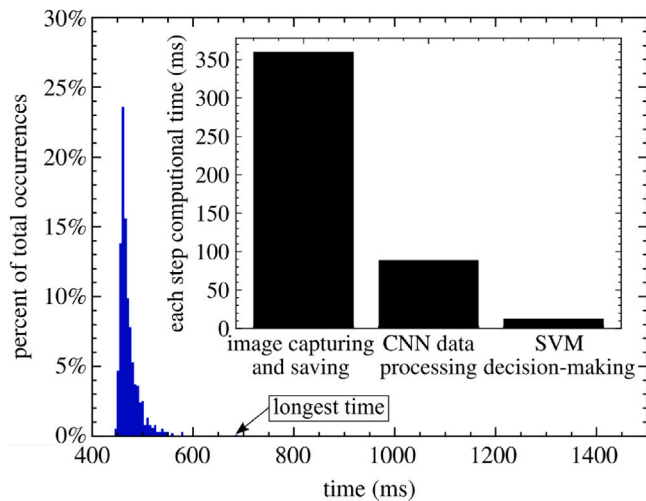
In these five incorrectly predicted samples, four miss classified ignorable cases were predicted to have impactful damage, and one impactful sample was incorrectly predicted as non-impactful. Considering impactful defects (the area outside the green decision-making boundary) as the positive class, samples 0384, 0387, 0396, and 409 are classified as false positives as the decision-making boundary incorrectly predicted these samples to have impactful damage cases. Sample 0388 was predicted to meet the tensile force requirement, but it did not, so it is a false negative error. Of the five incorrectly predicted samples, only two samples (0388 and 0409) did not meet the required tensile force and therefore are considered to be the more detrimental errors.

An important consideration is that samples 0384 and 0409 did not break at the seeded fault locations, suggesting that the detected defect was not the defect that limited their structural loading capability. It is hypothesized that sample 0384 was miss classified as it had the shortest defect length at 0.5, and the dataset does not have sufficient data at this defect size to accurately train a decision boundary. All five missed classifications are located near the decision boundary. As expected, samples located on or near the boundary are the most likely to be misclassified; this should be accounted for using the domain expertise of the practitioner when setting the load-reduction threshold used in building the decision-making boundary.

Table 2

Incorrect predictions for the 30 sample test data.

Sample number	Defect length	Defect location	Layers with defect	Tensile force achieved	Defect predicted as	Broken at seeded defect	Meet tensile force requirement (948.6 N)	Error type
0384	0.5 mm	10.4	2	953 N	impactful	No	Yes	false positive
0387	2.5 mm	6.08	3	950 N	impactful	Yes	Yes	false positive
0388	4.5 mm	5.10	1	941 N	ignorable	Yes	No	false negative
0396	1.8 mm	15.3	3	949 N	impactful	Yes	Yes	false positive
0409	5.5 mm	5.00	1	938 N	impactful	No	No	false positive

**Fig. 10.** Time range calculation for the real-time product structural quality validation.

3.3. Result of real-time defect detection and decision-making process

Results for online defect detection are shown in Fig. 10, which reports the recorded times for 1,000 cycles tested. The longest process time is 688.1 ms, and the mean processing time is 461.8 ms. As one layer printing time requires 75 s, this means the proposed product structural validation algorithm is feasible for real-time decision-making. Moreover, with a 100 times disparity between the required and available time, there is potential to deploy the proposed method on edge computing hardware [35].

To demonstrate the real-time defect detection and decision-making process, a specimen was designed with two defects during the printing process that is made up of four layers. The first and second layers are of good quality. Then a 5 mm defect at location 2.7 occurs on layer three, and another 15 mm defect is seeded at location 12.7. During the printing, as the first defect on layer three has a limited impact on product structural quality, the algorithm makes the decision to continue after evaluating the component health index. After finishing layer four, the new evaluation result shows that two defects stacked together undermine the product quality. Therefore, the algorithm displays a warning window to alert the user who may choose to cancel the print. Fig. 11 shows a screenshot of a video developed to demonstrate the defect detection and decision-making process in real-time. This video is included in the manuscript's supplemental material. This example demonstrates that the proposed real-time defect detection and decision-making system can make the right decision within a short time after a layer is finished.

4. Conclusions

This paper presented a real-time methodology for product structural validation for components manufactured by the material extrusion additive manufacturing (MEXAM) method. This work links real-time defect detection with product structural quality validation rather than

just focusing on defect detection. The specially designed material extrusion printer integrates an optical camera to obtain in situ images for training the convolutional neural network (CNN) model. The machine learning (ML) model detects damage and estimates every layer's quality. The output prediction from ML can be evaluated on the decision boundary made by the support vector machine (SVM) with the radial basis function kernel. Based on the evaluation results, the multi-dimensional accumulation-threshold-based decision-making system can decide whether to continue or cancel printing. The novel contribution of this method is its ability to link real-time defect detection with product structural quality validation.

The defect accumulation impact on different layers is considered, which makes the evaluation and validation process more comprehensive and precise. Results have shown that the developed product structural quality validation algorithm has good recall and F1 scores of 97.71% and 90.42% in a laboratory setting with the same image acquisition system used for training and validation. Further validation of the work requires that the algorithm be re-trained for more complex geometries and with different acquisition systems used for training and validation and the accuracy needs to be re-evaluated. Meanwhile, the longest and mean process times are 688.1 ms and 461.8 ms, which means the proposed algorithm satisfies the timing constraints for real-time product structural quality validation and decision-making. While the proposed real-time structural validation method achieved good results, one key drawback is that it is time and effort-consuming to obtain the decision boundary from the tensile test result. Future work will focus on automating and expanding the component health index to multiple dimensions and supplementing the training process with numerical simulations, then transfer and extend the approach to other additive manufacturing technologies. The dataset developed for this work is made available through a public repository [29].

CRedit authorship contribution statement

Yanzhou Fu: Writing – review & editing, Writing – original draft, Software, Methodology, Investigation. **Austin R.J. Downey:** Writing – review & editing, Methodology, Funding acquisition. **Lang Yuan:** Writing – review & editing, Methodology. **Hung-Tien Huang:** Software.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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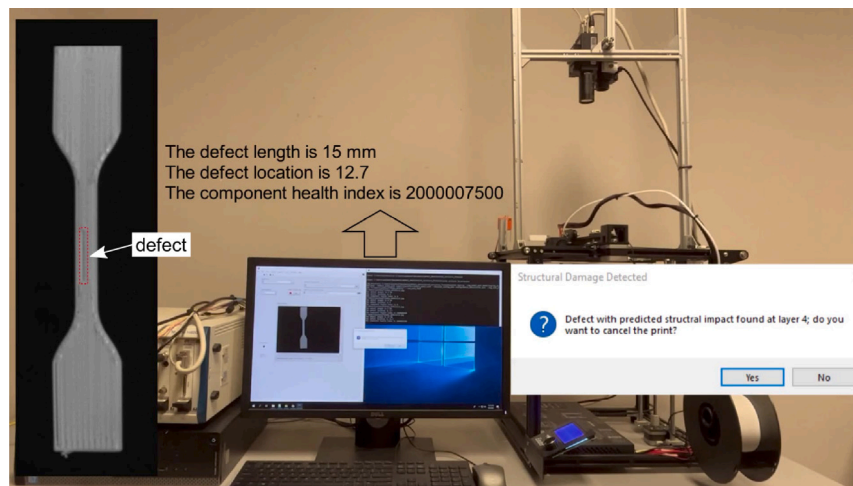


Fig. 11. Screenshot of a video demonstrating the defect detection and decision-making process in real-time, where the warning window appears when an impactful defect occurs. The video is available in the paper's supplemental documents.

Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.addma.2023.103409>.

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